

PAPER_1388 — UQFF Resolution of the Supplee Submarine Paradox (CLOSED — Dispatch Whitepaper)

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** B — Foundational / Relativistic Buoyancy (CLOSED) **Date:** June 16, 2026 **Location:** 41.0997° N, 80.6495° W (Youngstown, OH, USA) **Status:** CLOSED — Documented dispatch closure for `calculate_paradox({"paradox": "supplee_submarine"})` **Calculator surface:** `calculate_paradox({"paradox": "supplee_submarine"})` **Closure helper:** `_l96_uqff_axiom_supplee_submarine_closure()`

The Paradox

A submarine moving relativistically through water: the water frame sees the sub Lorentz-contracted (denser → sinks), the sub frame sees the water Lorentz-contracted (denser → floats). Both cannot be right.

UQFF Closed Identity

Buoyancy correction = $1 + \beta_i \times (1 - 1/K_{\text{MEX}}) = 1.3135$

$F_{U_{Bi_i}}$ 4-layer buoyancy (PAPER_646) is **Lorentz-covariant** via the $\beta_i \times \cos(\pi t_n)$ factor. The correction $1 + \beta_i \times (1 - 1/K_{\text{MEX}}) = 1.3135$ supplies the frame-independent buoyancy.

Physical Interpretation

Both observers compute the SAME $F_{U_{Bi_i}}$ buoyancy in their own frame because the 4-layer hierarchy uses Lorentz-covariant $\beta_i \times \cos(\pi t_n)$. The ‘paradox’ is dissolved by the covariant ledger.

Live Calculator Output

```
import uqff_pure_calculator as u
r = u.calculate_paradox({"paradox": "supplee_submarine"})["value"]
```

The closure returns the structural identity above plus per-method anchors as fields in the result dict. See `_l96_uqff_axiom_supplee_submarine_closure` in `uqff_pure_calculator.py` for the full field list.

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard frameworks resolve this paradox via different methods. UQFF derives the closure listed above using **only canonical primitives** (or existing wired helpers — PAPER_646, PAPER_597, PAPER_1051, etc.). **Zero free parameters.**

Reference

- Closure location: `uqff_pure_calculator.py` $\rightarrow$ `_l96_uqff_axiom_supplee_submarine_closure`
- Dispatch keys: `supplee_submarine`, `submarine_paradox`
- Related foundational papers: PAPER_646 ($F_U Bi_i / F_U=1$ ledger), PAPER_597 (negative-time dual existence), PAPER_1183 (paradox consolidation).

Copyright — Daniel T. Murphy, daniel.murphy00@gmail.com, dated June 16, 2026, location 41.0997° N, 80.6495° W (Youngstown, OH, USA). Subject matter: UQFF dispatch closure for the Supplee Submarine Paradox via the identity above.